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Abstract

Social media has risen within the past 15 years to become a pervasive influence on the lives of the majority of people in 2020. Previous research in the field has focussed on the potentially addictive nature of high levels of social media use. So far results regarding the cognitive effects of intense social media use have not been conclusive therefore this study attempts to investigate if there is measurable attentional bias for social media related stimuli across a range of social media users. A representative sample of 82 social media users (41 female/39 male, 18-30 years) completed a Stroop task which assessed reaction time for social media related words and neutral words. In addition participants completed a series of established self-report questionnaires to assess social media, smartphone and general technology related behaviours and attitudes. Finally participants were asked to provide iPhone screen time data in relation to their smartphone and social media use. Users who identified as high users of social media were found to have no significant bias for social media related words, similar to low and medium social media users. In addition high social media users who also exhibited high smartphone use showed no significant bias for social media related words. Due to the high exposure to social media across 99% of the targeted demographic it is possible familiarity contributed to the lack of results. Research on social media use and its effects on cognitive processes remain inconclusive, understudied and rarely objective. We propose there must be more urgent focus on understanding the potential cognitive (and social) effects of intense social media use.

Social media refers to virtual communities where users can interact with real-life and online friends, create pages or profiles and interact with others based on shared interests (Kuss and Griffiths, 2011). Social media began back in the 1990s with the launch of SixDegrees, which allowed users to upload profiles and make friends with other users. An exponential rise in social media use occurred with the creation of MySpace in 2003 and the advent of Facebook being made publicly available in 2007. Around the same time as Facebook we saw the emergence of the smartphone with Apples iPhone releasing in 2007. This rise in social media users is at least partially supported by Griffiths 2003 "availability hypothesis", roughly translating to more opportunity to engage in social media due to increased access via smartphones. In turn this 'availability' leads to in an increasing number of people engaging in both activities, using smartphones and social media. The convergence of always-on technology (smartphones) and instant networking (social media) presents many concerns at global, societal and individual levels, yet we still know rather little about the potential effects of the pervasive nature of social media (enabled by smartphones) on human cognitive processes.

Availability and pervasiveness of social media

In the United Kingdom in 2020 social media has an overall population penetration rate of 66% (45 million users) and the most commonly used social media networks in the United Kingdom during May 2020 are YouTube, Facebook, Whatsapp, Messenger, Instagram and Twitter, as shown in Table 1.

Table 1 Social media penetration in the United Kingdom, May 2020 Source: statista.com

Social media penetration in the U.K. increases by 2-10% every year, adding urgency to the need for further research on the impact social media networks have on human cognition. In addition the use of

smartphones as the only way of accessing the internet and social media sites in increasing – from 8% in 2017 to 11% in 2018 (Ofcom, 2019), with computer use falling by a similar percentage. For those in their twenties a fifth use only their smartphone to go online and 77% of 16-24 year olds say they would miss their smartphones most of all their devices.

Given the almost-complete global penetration of social media, its power (increasingly enabled by smartphones) to connect, influence and mobilise people across the world is beyond our current experience or description.

Social media and young people

Of particular concern is the effect of social media on adolescents and young people. Those born after 1995 had a very different childhood due to the influence of digital technology like the world wide web, home computers, tablets and smartphones. Ofcom (2017) found that 99% of British 12-15 year olds go online daily and 83% have their own smartphone. In 2019 Ofcom reported that 99% of 18-24 year olds had a smartphone and 93% have at least one social media profile. Most have more than one social media profile across different networks, using each network for a different purpose. By age 25-34 there is a small drop in the number who have multiple social media profiles (91%) while smartphone ownership remains at 99%. Overall less than 1% of British young people and adolescents are not engaged with social media.

For young people social media has become essential to forming friendships, gaining social status and support and for the development of individual attitudes, behaviours, goals and identity. The essentialness of young people using social media usually stems from peer pressure (Borca, Bina, Keller, Gilbert, and Begotti, 2015; Poulin and Chan, 2010) and they typically perform a wider range of activities than older adults (Ofcom, 2019). Borca et al. (2015) went on to explore how young people used social networks to identify with new identities, particularly those young people who tended towards introversion or social anxiety. Throuvala, Griffiths, Rennoldson and Kussa (2018) described how adolescents see social media as indispensable to a range of social interactions and relationships, a way to get information, validation and place of inspiration and learning. They also reported a “symbiotic relationship with peers online via social media and smartphone attachment” (Throuvala et al., 2018 p168) include FoMo (fear of missing out) as a key driver of young people’s social media use. Young people are sophisticated social media users, assigning different purpose to different platforms and actively using advanced rarely used features, such as Finstagram on Instagram to ensure privacy and security. For this generation Facebook is the least favourable platform due to its perceived complex features and it being thought of as their parents platform of choice.

Young people also more commonly consider themselves ‘addicted’ to social media according to a literature review by Cabral in 2011. Cabral’s results, almost a decade ago, showed that young people were worried about the dominance of social media in their lives and how it affected their ability to attend to other aspects of life such as school, work and real-life relationship. Although young people recognised a need to decrease their usage of social media they continued to make it their top priority, possibly related to the social pressure, peer group support and FOMO referred to previously. Using Griffiths components of non-chemical addictions it appeared from this early study that social media use pointed towards addictive behaviours with specific reference to tolerance, salience and relapse.

Technological déjà vu

Most studies into social medias effect on our cognitive processes appear to still focus on the potentially addictive, problematic or pathological effects of social media. Similar concerns have been expressed with the emergence of other technology-related changes in the past, for example televisions, computers, gaming devices and smartphones. Davidson and Ellis (2019) called this ‘technological déjà vu’ and described the ‘moral panic’ around new technologies as always producing negatively focused research. This is noticeable in the questionnaires used to measure social media and smartphone use, which are often based in depression, anxiety or addiction. Another example of this tendency to seek problematic outcomes is evident in the collection of specific mental health disorders attributed to use of newer technologies - for example ‘nomophobia’ (Yildirim and Correia, 2015) which describes a

persistent form of separation anxiety caused by a fear of not having their mobile phone and 'cyberchondria' (Lewis, 2006) relating to the phenomenon of feeling ill after consuming online health information, which he refers to as 'cyberquackery'.

Measuring the effects of social media

As mentioned, research so far has focussed on building addiction based diagnostic models of problematic social media (or smartphone) use. As early as 2001 Davis developed the Pathological Internet Use (PIU) model which considered not only urges, tolerance and withdrawal of the internet but also the social aspect of internet use. Davis considered pathological use of the internet a mechanism to provide support that for some reason isn't available in real-life. In 2016 Müller et al. focused on drawing parallels between the DSM-recognised Internet Gaming Disorder and use of social networking sites. They found that high use of social networking correlated to higher levels of addiction criteria and also to "psychosocial distress". The Social Media Disorder Scale was developed by van den Eijnden, Lemmens and Valkenburg in 2016, again based on the criteria for Internet Gaming Disorder. They sought to provide a clear diagnostic tool to distinguish addicted users from those who were simply high engagers using a relatively large sample of adolescent boys. They found a moderate positive relationship between daily social media use and attention deficits. Wegmann and Brand (2020) also drew on Internet Gaming Disorder to compare cognitive correlates for potential social network use disorder. However they came to the conclusion that there is a dearth of research on the cognitive aspects of intense social media use and therefore no conclusions were possible just yet. Balakrishnan and Griffiths (2017) even developed a scale to measure a condition known as "selfitis", proposed as a mental disorder by the American Psychological Society. It describes the obsessive taking of 'selfie' photos and posting them on social media, something only possible due to the convergent emergence of smartphones, enabling digital camera functionality and social media network sites, facilitating the sharing and rating of the photographs. Brailovskala and Bierhoff (2018) correlated this 'self-presentation' behaviour to higher levels of narcissism within the Millennial generation.

Across most of these studies there are issues with bias, reliability and ecological validity. And ultimately the key question remains of causality - do people use social media to compensate for or facilitate their mental disorder or does using social media cause the disorder?

Although addiction is a the most common basis for research into social media use, others have also considered associations between high social media use and ADHD. For example, Boer, Stevens, Finkenauer, and van den Eijnden (2019) in a rare, albeit short term, longitudinal study found the intensity of social media use had no effect on increasing ADHD symptoms over time. They did however find that a high number of social media problems, as measured by the Social Media Disorder Scale, did increase ADHD symptoms over time. This finding differentiated effects between the intensity of social media use and the behaviours that surround how social media is used.

In 2017 Primack et al. considered the use of multiple social media platforms and symptoms of depression and anxiety, finding that time spent on social media was less significant than the number of social media platforms a user was active on when considering increase symptoms of depression and anxiety. Primack found a moderate correlation between the propensity towards multi-tasking across social media networks and a decreased attentional ability. Twenge, Martin and Campbell (2018) revealed lower levels of non-psychological well-being among adolescents who spent more time on computer technologies, i.e. with higher screen times. Those who spent the most time on screens took part in less non-screen activities too. There is a possibility that some findings in psychological studies, which are attributed to social media alone are instead related, or act in addition, to the increased screen time necessary to indulge in social media networking behaviours.

A 2019 study by Nikolaidou, Fraser and Hinveest used a dot probe task to investigate attentional bias amongst individuals who were identified as problematic social network users. They found that problematic social network users showed an attentional bias towards social media related images compared to control images. They came to the conclusion that problematic social media use, as self-reported using addiction-based questionnaires, was similar to known addiction mechanisms. Given the underlying factors of the dot probe task and problematic usage questionnaires was addiction we use a modified design in this study that aimed to avoid measuring similar constructs.

While research on social media (and smartphone) use focusses on addiction and other concerning issues there are at least a few studies which take an objective approach to considering the consequences on cognitive processes of those who display a higher than normal use of social media. There are some heartening findings in the midst of the seeming focus on finding problems. In 2015 Barr, Pennycook, Stolz and Fugelsang found no significant difference in cognitive measures between smartphone users and those who didn't use smartphones, nor did they find differences between levels of smartphone use amongst users. Borca et al. (2015) concluded that parents who typically worry about deemed excessive internet and social media use need to be more open to the benefits - that technology allows young people to develop knowledge, self-identity and autonomy. Ellis, Davidson, Shaw and Geyer (2019) argued that increased facilitation of social interactions and stable social support environments provided positive health benefits as found in previous studies on social networks and smartphone use (Haslam et al., 2018; Pachucki, Ozer, Barrat, and Cattuto, 2015). Ellis (2019) proposed a range of possible benefits due to new technologies like smartphones, particularly in relation to the facilitation of social relationships and support. Add to this, in 2019 Orben's narrative review concluded that there were no concrete effects on well-being from digital technologies, such as smartphones and social media, although in part due to low quality research and other limitations. In the same year Zendle and Bowden-Jones (2019) argued that we must practice caution when assigning an addictive quality to social media use, pointing out most research is based on self-reported measures and there is no clear-cut parallel with other addictive mechanisms such as substance or gambling addictions.

Again it's important to consider Davidsons technological *déjà vu* when reviewing available research –social media and smartphones have become such an integral part of our lives we often forget both are but a mere 13 or so years old.

Challenges when measuring the effects of social media

One persistent challenge when measuring social media use is that psychological research has almost exclusively used a range of self-report questionnaires to measure it, focussing on personal experience instead of measuring the actual behaviours. Yet today we can use several methods to collect data 'in situ' as users engage with social media in their natural environment, thereby increasing ecological validity of future studies. The simplest method is to collect data from smartphones instead of relying on estimates, this approach can be used to provide information on general smartphone use and more detailed social media use and behaviours. In 2018 Gower and Moreno utilised screenshots of iPhone battery as an experiment to improve reliability of smartphone data, using the ability to collect actual smartphone data instead of relying on biased self-assessments. Preceding this back in 2015 Andrews, Ellis, Shaw and Piwek developed an app to collect smartphone data and diaries and found that smartphone users self-reported use were moderately related to their actual behaviours. In 2018 Giunchiglia, Zeni, Gobbi, Bignotti and Bison used a similar approach, using an app called iLog in combination with a diary to predict academic performance – they found using social media during academic activities had a negative effect, less so with high levels of general social media use. Loida, Tähta, and Rozgonjuka (2019) looked specifically at providing a usage pop up notification on smartphones and how the provision of that information affected problematic use and other general smartphone behaviours. They found that providing such information had no effect on self-reported problematic smartphone usage, nor did it effect screen time or how often participants checked their phone. Ellis et al. (2019) later investigated if smartphone scales predicted actual behaviours and concluded that the available self-report scales were not sensitive enough to predict technology usage or behaviours and that conclusions reached using them were psychologically unsound. For this reason our study captures iPhone screen, app, notification and pickup data in addition to the self-report questionnaires.

Another challenge when measuring social media use is the rapid proliferation, ascent and descent of social media network brands. Much of the available research on social medias effect on our cognition focuses on Facebook use. Although in 2020 Facebook remains the world's most popular social media network (today it has 2.5 billion active monthly users worldwide (<https://www.statista.com/>) as shown in

Figure 1) other social networks are almost always used in addition to, and more often than, Facebook.

Figure 1 Active worldwide users of social media networks in 2020 (millions)

Facebook is also no longer the most popular social media network for young people specifically, with YouTube, Instagram and Snapchat being in the lead (Ofcom, 2019). In 2020 social media has been expanded to include not only social networking sites like Facebook but also content-sharing sites like YouTube, Instagram and Snapchat. As technology becomes more agile it becomes more challenging to measure the different facets of usage, behaviour and attitudes across the numerous social media platforms.

Why attention is important

Our study attempts to measure the effects of high levels of social media use on selective attention. Attention is a key cognitive process for our everyday life where we are constantly exposed to multiple environment stimuli. Considering the saturation and always-on availability of social media there remain questions as to how social media use affects our ability to attend to other things in our environments. We read stories in the press of people dying because they were taking selfies to post on social media or of people being killed crossing the road while looking at their smartphone without raising their eyes.

Attentional bias describes the phenomenon of our tendency to selectively attend to certain stimuli over others and is used frequently to measure addictive tendencies. Robinson and Berridge's (1993) incentive-sensitization theory posits that related stimuli have an 'enhanced saliency', i.e. relevancy to the individual, so these stimuli are processed while selective attention causes other categories of stimuli to be disregarded or ignored. Classical conditioning through repeated exposure means the related stimuli "grabs attention, becomes attractive and wanted and thus guides behaviour to the incentive" (Robinson and Berridge, 1993, p. 261), creating a subjective craving, or desire and preoccupation (Franken, 2003), for the related stimuli which in turn increases selective attention bias – reinforcing the repeated behaviour.

This study used a modified version of the most popular way to test stimuli-related attentional bias, i.e. the Stroop task (Stroop, 1935) based on another modified version by Cox et al., 2006. Selective attention is measured using two categories of words, neutral and words related to the salient stimuli. The words in both lists are matched on etymological features such as phonemes, length and frequency of occurrence in language to control for any latencies between the salient and neutral words. Trials are presented as single words displayed in any one of several colours (usually red, yellow, blue, green). The task is to name the colour as accurately and quickly as possible. Where an attentional bias exists the assumption is that it will take longer to identify the colour as the participants attention is grabbed by the semantic context. Sometimes a simple index of attentional bias is calculated as the difference between the mean reaction time for the categories of words.

Aims and hypotheses

Given the focus on the potentially addictive nature of social media remains inconclusive this study aims to separate diagnosis of a disorder from a study of the effect social media has on cognitive function. We used both self-report and objective measures, i.e. iPhone data, to determine if there is any relationship between social media use, smartphone use and selective-attention processes.

In summary, the aim of this study was to objectively investigate attentional bias for social media related words in relation to the intensity of social media and smartphone use. Our hypotheses were: (1) high social media users would be slower to react to social media related words than those participants with lower rates of social media use, thereby displaying attentional bias for social media related stimuli and (2) those who were high users of both social media and smartphones would display an even greater attentional bias for social media related stimuli.

Method

Participants

Participants were recruited from the students attending the University of Strathclyde via email and online forums and also from the general public via social media advertisements. Some participants were recruited using Prolific, a platform which pays a small sum of £2.50 for each survey completion. All participants were aged 18-30, were iPhone users and had normal or corrected-normal visual acuity without colour vision deficiency or blindness. Participants who failed to complete the task and questionnaire were discarded (19), leaving a group of 88 participants. Further analysis of outliers resulted a participant group size of 82.

Demographic information was designed according to the standards set by the Office for National Statistics (2015). The group consisted of 41 Females, 39 Males and 2 Prefer not to say/Other with a mean age of 22.73 (3.50), see Table 1 for more information.

Table 1 Demographic characteristics (N=82)

Design

The independent variables for the main study were the mean scores of the Social Media and Smartphone subscales from MTUAS. The dependant variables were the mean reaction times for social media related words and neutral words during the Stroop task. The study used a cross-sectional design to 'take a snapshot' of population groups which enabled further investigation of characteristics such as age, ethnicity, gender, etc.

Materials

Twenty-five neutral words and 25 social media related words were used in the Stroop task (stimuli are listed in the Appendix). When considering which social media networks to use to generate stimuli we excluded those that are generally used for purposes other than social networking. This eliminated YouTube (video), Whatsapp(phone), FB Messenger(FaceBook specific phone/chat). For social media related words, co-locates for "Facebook", "Twitter" and "Instagram" were generated from the iWeb Corpora (<https://www.english-corpora.org/iweb/>) and Corpus of Global Web-Based English (GloWbE) (<https://www.english-corpora.org/glowbe/>). The top 50 co-locates for each social media network word for both corporas were then analysed to produce a merged 25 top co-locates which resulted in the social media stimuli. Neutral words based on common household items were identified using a similar process and then the sets analysed individually and as a paired list (<http://psico.fcep.urv.cat/utilitats/nim/index.php>) to ensure frequencies did not differ across both sets of words [$t(20) = -0.13$, $p = 0.90$]. It was not possible to do this for social-media related words "instagram", "facebook" and "hashtag".

The Stroop Task was hosted on Inquisit and administered online with participants using Inquisit Player v6.1.

SUQ

The Smartphone Usage Questionnaire was developed by Marty-Dugas, Ralph, Oakman and Smilek in 2018 in order to measure the relationship between smartphone use and everyday inattention. It comprises two parts - the Smartphone Use Questionnaire: General (SUQ-G) and the Smartphone Use Questionnaire: Absent-Minded (SUQ-A). Each contains 10 items with responses measured on a 7-point likert scale, ranging from 1 (Never) to 7 (All the time). The SUQ-G measures the frequency of smartphone-related behaviours, for example how often they use social media, browse the web or engage in text messaging and has shown strong internal consistency (Cronbach's alpha = 0.94). The SUQ-A assesses the frequency of absent-minded smartphone behaviours has shown strong internal consistency (Cronbach's alpha = 0.83) and is cited across similar studies although repeatability of significant results is an issue for at least one recent student research project (not yet published).

MTUAS

In 2013 Rosen, Whaling, Carrier, Cheever, and Rokkum developed the Media and Technology Usage and Attitude Scale, attempting to overcome the emergence of new technologies and the tendency for researchers to focus on only one element of technology use. MTUAS is a well-cited 60-item questionnaire, 44 items measuring technology usage and 16 measuring attitudes towards technology. Demographic questions were modified to accommodate British norms, using the Office for National Statistics Harmonised Concepts and Questions for Demographic information. The Usage scale consists of 11 subscales (smartphone usage, general social media usage, Internet searching, e-mailing, media sharing, text messaging, video gaming, online friendships, Facebook friendships, phone calling). For the purposes of this study online friendships and Facebook friendships were merged to create a subscale of 4, focussed on the participants self reported most used social media network. Smartphone and social media subscales account for the highest variance within the MTUAS, 11.94% and 11.61% respectively and were utilised as predictive variables for analysis. The social media behaviours subscale measures the frequency of social media related behaviours, for example how often they check their social media, post status updates or click "Like" on posts. The smartphone subscale measures the frequency of smartphone-related behaviours, for example how often they listen to music on the phone, make and receive text messages and take pictures using the phone. The overall Usage-based subscale assesses everyday usage of a broad range of technologies. The Attitude-based subscales measured negative attitudes, positive attitudes, anxiety and/or dependence and task-switching preference. Each subscale can be used together or separately as they have been proven as internally reliable and externally valid.

iPhone Data

Participants were asked to refer to their iPhone Screen Time data at this point, which asked them to enter their daily average screen time, weekly average social media time, daily average notification and daily average pickups. Due to an unexpected issue in how iPhone collects weekly social media time this data was ignored for the purposes of this study.

Procedure

This study, including the experimental protocol, was approved by the Ethical Committee of the University of Strathclyde. Initially participants completed an informed consent form and then completed the Stroop Task. All participants received all trials. Each trial in the Stroop Task started with instructions followed by 12 practice trials (excluded from statistical analysis) then 200 trials (100 social media trials and 100 neutral trials). Each word was presented 4 times, once in each colour, in a random order for each participant. Words were presented in uppercase, in the centre of a grey screen. Participants were asked to respond to the colour of the word as fast and as accurately as possible by pressing a keyboard button. At the top of the screen the corresponding colours ("red", "green", "blue" and "yellow") and keys ("d", "f", "j" and "k") were shown throughout the task. Each trial began with a 500ms blank screen and then a word was presented. There was no limit to response time and the word remained on-screen until a response was given. The two types of words were presented in separate blocks, that is, a set of social media related words and another set of neutral words as blocked presentation of stimuli are more likely to produce a significantly larger effect size when compared to randomized presentations (Holle, Nelly, and Heimberg, 1997, Bar-Haim, Lamy, Pergamin, Bakermans-Kranenburg, and van IJzendoorn, 2007). After completing the task participants completed the questionnaires and entered their iPhone data. They were then debriefed and asked for a final time for consent to use their data.

Results

Raw data was downloaded from Qualtrics and exported to excel. Reaction time data from filler trials and from trials with errors were removed. Data which was extensively incomplete was discarded. Five participants had outlying high error rates (at least 50%), as surmised from a box and whisker plot and were discarded from the study. For the remaining participants, the mean percentage of errors was <4%. To deal with extreme trial outliers, reaction times were excluded if they were evident on a box and whisker plot. 1.86% of all trials were discarded (2.05% for the High group, 1.95% for the Medium group and 1.55% for the Low Group). Means were calculated for correct reaction times for both social media related words and neutral words. Mean Social Media use as measured by the Media and Technology Usage and Attitude Questionnaire was 5.70 (1.61, range 1.67-10) on a 1-10 point scale, with higher values indicating more time spent on social media. Mean Smartphone use as measured by the Media and Technology Usage and Attitude Questionnaire was 6.35 (1.28, range 2-10) on a 1-10 point scale, with higher values indicating higher levels of smartphone use. Both skewness and kurtosis were acceptable. Descriptive statistics for MTUAS Social Media Use and Smartphone Use subscales as shown in Table 2.

Table 2 Descriptive statistics for predictive variables.

Participants were assigned to a group "Social Media Use" according to their MTUAS Social Media score using a tertiary split labelled "Low", "Medium" and "High". A tertiary split was chosen as it's considered more effective than a binary or median split (Gelman and Park, 2008). Participants were assigned to a second group "Smartphone Use" based on their MTUAS Smartphone score, however a tertiary split failed to provide the necessary normality nor homogeneity and was discarded. We proceeded with a median split of "Low" and "High" for "Smartphone Use". Group means are shown in Figure 2.

Figure 2 Mean RTs by Social Media and Smartphone Use

Normality tests were significant using Shapiro-Wilk across the board, however Levene's showed there was homogeneity of variance so we proceeded with a 3 (social media use: low, medium, high) $\times$ 2 (smartphone use: low, high) $\times$ 2 (social media related words, neutral words) mixed ANOVA. Stroop reaction times to different types of stimuli (neutral, social media related) were the within-subjects variable and social media use and smartphone use as the between-subjects' factors. We had predicted that participants in the high social media usage group would be slower to respond to social media related words than neutral words when compared to the low and medium use groups, however according to our results the social media group $\times$ stimuli interaction was non-significant [$F(2, 76) = 0.82$, $p = 0.44$], with critical F calculated as 3.12, p set at 0.05.

In addition we predicted the interaction of smartphone use and social media use would result in slower RTs for social media related words. This too was shown to be non-significant at $F(2, 76) = 0.04$, $p = 0.96$, with a critical F value of 3.12. There were no significant main effects. To check whether our non-significant results were due to a lack of statistical power, we conducted post hoc power analyses using G*Power 3 (Faul, Erdfelder, Lang, and Buchner, 2007) with power ($1 - \beta$) set at 0.80 and $\alpha = .05$. This showed us that sample sizes would need to be $N = 70$ in order for group differences to reach statistical significance at the .05 level, therefore it is unlikely our negative findings are related to a limited sample size, given $N = 82$. In summary, we concluded that neither high levels of social media use nor smartphone use (based on the self-reported MTUAS) had a significant effect on reaction times during the Stroop task and therefore showed no attentional bias effect for social media related words.

Further correlational analysis between self-reported measures and actual iPhone data was carried out, see Table 3. We did not find significant positive correlations between iPhone screen time minutes and MTUAS Smartphone Use, however there was a strong correlation between MTUAS Social Media Use and iPhone Screen Time, number of notification and number of pickups.

Table 3 Correlations between MTUAS and iPhone data

** Correlation is significant at the 0.01 level (2-tailed).

* Correlation is significant at the 0.05 level (2-tailed).

Given the lack of significant correlation between actual iPhone minutes, notifications and pickups and MTUAS Smartphone use we questioned whether the MTUAS Smartphone scaled was reliable. To explore the relationship between Social Media Use and iPhone data further we carried out a further

mixed ANOVA using Social Media Group and actual iPhone screen time.

In our original prediction we wanted to test if the interaction of smartphone use and social media use would result in slower RTs for social media related words. Using a median split on iPhone minutes instead of the self-reported smartphone use measure we again found normality tests were significant using Shapiro-Wilk across the board, however Levene's was not significant so we proceeded with a 3 (social media use: low, medium, high) $\times$ 2 (iPhone minutes: low, high) $\times$ 2 (social media related words, neutral words) mixed ANOVA. This interaction of social media use and actual iPhone minutes was also shown to be non-significant at $F(2, 76) = 1.02$, $p = 0.37$, with a critical F value of 3.12, in addition no main effects were found.

Overall our analyses showed that high levels of social media use, even when combined with high levels of smartphone use – both self-report and actual iPhone usage - did not have a detectable effect on attentional bias as measured by the Stroop task.

Discussion

This study concluded that high levels of social media use had no effect on attentional bias for social media related words compared to neutral words. Furthermore we found that social media and smartphone use together – self reported or actual – do not have significant impact on reaction time.

These results are contradictory to our hypotheses 1) that high levels of social media use would result in a delayed ability to react to social media related words during the Stroop task as compared to participants with lower levels of social media use and 2) that high levels of social media use, combined with high levels of smartphone use, would result in an even longer delayed ability to react to social media related words during the Stroop task as compared to all other participants.

Our results put into question previous theories of the effect of social media on selective attention and addiction. For example, Nikoladiou et al. used a dot probe experiment to conclude there was attentional bias to social media related stimuli, however in their study participants were categorised by self-reported problematic social media use and not merely grouped by their use of social media. There are few comparable studies (self-reported and actual data measures with an experimental component) and most that are available are focused on problematic use. When considering our analysis we found few studies with similar hypotheses. One similar study which does bear out similar results is Barr et al.'s 2015 research into the effects on attention and levels of smartphone use, where they found no effect on attentional bias when considering levels of smartphone use. Ellis (2019) proposes that existing studies provide poor-quality evidence and poor definition of problematic use.

There are several limitations to our own study. Our participants comprised a self-selected sample of convenience and resulted in a mix of demographics that is unlikely to generalise across the population. Our sample was mainly white, single students with little range across income and living situation. In addition numerous co-variables have been identified in previous research that may affect results, for example Müller et al. (2016) found personality factors played a part in predicting the effects of intense social media use.

The Stroop task relies on a level of familiarity with the salient stimuli, however given the lack of a control group when it comes to social media users, i.e. people who do not use social media, it's entirely possible that there is a limitation in what can be measured from the Stroop as the differences between low and high levels of use aren't enough to affect familiarity. We based our Stroop on the three most popular social networks – Facebook, Instagram and Twitter in order to manage the social media related words for maximum coverage. However from our data there are a number of participants who did not use any of those networks frequently. The number, and varying purposes, of social media networks makes it challenging to design an appropriate experiment to cover the potential differences in a populations favoured social media networks. In future studies using Stroop for social media purposes it would be useful to either limit participants to frequent users of specific networks or to create a dynamic design that allows the capture of individuals favoured networks and ensuing customised collection of data. This could potentially allow analyses between social media groups, for example comparing attention bias between Facebook and Twitter users.

Etymologically, future Stroop studies would require the ability to match phenomes of social media related words, this is not yet an available resource beyond everyday language therefore this study was unable to match words confidently to leverage the full effects of Stroop.

This study used two self-report questionnaires to measure social media and smartphone use. Self-report is limited to personal experience and offers little objective data on actual use or behaviours. This was reiterated by our failure to correlate the MTUAS Smartphone subscale to actual iPhone minutes, number of notification or number of pickups. Collecting iPhone social media data did provide not easy way to assess intensity of social media use. Due to time-zone differences and participants participating on different days at different times (the week figures begin on a Monday at 00:00:00) it was not possible to compare their screen time on social media alone. Zendle and Bowden-Jones (2019) suggest that using industry data will be the most effective way to understand “how problematic the excessive use of social media is”. We would argue that social media and smartphone industries data would be critical to further facilitate longitudinal research across all disciplines that seek to understand the human form, psyche or social self. We highly recommend that future research take advantage of technology to improve the ecological validity of social media use, behaviours and attitudes.

Currently there is a strong trend within technology to utilise psychology and social science alongside computer and data science to strategically maximise usage of social media platforms and physical technologies such as smartphones and tablets. For example, Van Meter, Syrdal, Powell-Mantel, Grisaffe and Nesson (2018) proposes that attitudes towards social media are of less value to marketing than attachment. This bears out in the rise of newer technology disciplines dedicated to engaging users emotionally through brand and functionality and the variety of media and channels emerging from newer social media networks. Yet there remains little human-centric research available to balance the needs and aims of the technology industry with that which makes us human. Understanding how our new digital lives are shaped by the technology all around us is key to ensuring not only commercial interests drive the symbiotic relationship between man and machine.

In summary, the results of this study, albeit with some limitations, showed that neither intensity of social media use nor smartphone use created a significant attentional bias towards social media related words.

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List of Appendices

Appendix A

Smartphone Usage Questionnaire (SUQ)

There are two parts to the SUQ. SUQ-G measures the frequency with which participants engaged in a broad range of smartphone-related behaviours, such as how often they send and receive text messages, use social media, or browse the web on their smartphone. SUQ-A was designed to assess the frequency with which individuals engaged their phones absent-mindedly

The following statements are about smartphone usage and certain experiences that you may have while using your smartphone. We are interested in how frequently you have these experiences on a typical day.

SUQ-G1. How often do you have your cellphone on your person?

|_____|

1. Never 7 All the Time

SUQ-G2. How frequently do you send and receive text messages or emails?

|_____|

1 Never 7 All the Time

SUQ-G3. To what extent do you have push notifications enabled on your phone?

|_____|

1 Never 7 All the Time

SUQ-G4. How often do you find yourself checking your phone for new events such as text messages or emails?

|_____|

1 Never 7 All the Time

SUQ-G5. How often do you use the phone for reading the news or browsing the web?

|_____|

1 Never 7 All the Time

SUQ-G6. How often do you use sound notifications on your phone?

|_____|

1 Never 7 All the Time

SUQ-G7. When you get a notification on your phone, how often do you check it immediately?

|_____|

1 Never 7 All the Time

SUQ-G8. How often do you use the calendar (or similar productivity apps?)

1 Never 7 All the Time

SUQ-G9. How often do you check social media apps such as snapchat, facebook, or twitter?

1 Never 7 All the Time

SUQ-G10. How often do you use your phone for entertainment purposes (i.e. apps and games)?

1 Never 7 All the Time

SUQ-A1. How often do you open your phone to do one thing and wind up doing something else without realizing it?

1 Never 7 All the Time

SUQ-A2. How often do you check your phone while interacting with other people (i.e. during conversation)?

1 Never 7 All the Time

SUQ-A3. How often do you find yourself checking your phone "for no good reason"?

1 Never 7 All the Time

SUQ-A4. How often do you automatically check your phone without a purpose?

1 Never 7 All the Time

SUQ-A5. How often do you check your phone out of habit?

1 Never 7 All the Time

SUQ-A6. How often do you find yourself checking your phone without realizing why you did it?

1 Never 7 All the Time

SUQ-A7. How often have you realized you checked your phone only after you have already been using it?

1 Never 7 All the Time

SUQ-A8. How often do you find yourself using your phone absent-mindedly?

1 Never 7 All the Time

SUQ-A9. How often do you wind up using your phone for longer than you intended to?

1 Never 7 All the Time

SUQ-A10. How often do you lose track of time while using your phone?

|-----|

1 Never 7 All the Time

Appendix B

Media and Technology Usage and Attitudes Scale

Usage. subscales

This scale includes 44 items which comprise 11 subscales: Smartphone Usage (9 items), General Social Media Usage (9 items), Internet Searching (4 items), E-Mailing (4 items), Media Sharing (4 items), Text Messaging (4 items), Video Gaming (3 items), Online Friendships (4 items), Online Friendships (2 items), Phone Calling (2 items) and TV Viewing (2 items)

10-point frequency scale for items 1–40 (with scoring in parentheses):

Never (1)

Once a month (2)

Several times a month (3)

Once a week (4)

Several times a week (5)

Once a day (6)

Several times a day (7)

Once an hour (8)

Several times an hour (9)

All the time (10)

Please indicate how often you do each of the following e-mail activities on any device (mobile phone, laptop, desktop, etc.)

(E-mailing subscale) Send, receive and read e-mails (not including spam or junk mail).

(E-mailing subscale) Check your personal e-mail.

(E-mailing subscale) Check your work or school e-mail.

(E-mailing subscale) Send or receive files via e-mail.

Please indicate how often you do each of the following activities on your mobile phone.

(Text messaging subscale) Send and receive text messages on a mobile phone.

(Phone calling subscale) Make and receive mobile phone calls.

(Text messaging subscale) Check for text messages on a mobile phone.

(Phone calling subscale) Check for voice calls on a mobile phone.

(Smartphone usage subscale) Read e-mail on a mobile phone.

(Smartphone usage subscale) Get directions or use GPS on a mobile phone.

(Smartphone usage subscale) Browse the web on a mobile phone.

(Smartphone usage subscale) Listen to music on a mobile phone.

(Smartphone usage subscale) Take pictures using a mobile phone.

(Smartphone usage subscale) Check the news on a mobile phone.

(Smartphone usage subscale) Record video on a mobile phone.

(Smartphone usage subscale) Use apps (for any purpose) on a mobile phone.

(Smartphone usage subscale) Search for information with a mobile phone.

(Text messaging subscale) Use your mobile phone during class or work time.

(TV viewing subscale) Watch TV shows, movies, etc. on a TV set.

(TV viewing subscale) Watch video clips on a TV set.

(Media sharing subscale) Watch TV shows, movies, etc. on a computer.

(Media sharing subscale) Watch video clips on a computer.

(Media sharing subscale) Download media files from other people on a computer.

(Media sharing subscale) Share your own media files on a computer.

(Internet searching subscale) Search the Internet for news on any device.

(Internet searching subscale) Search the Internet for information on any device.

(Internet Searching Subscale) Search the Internet for videos on any device.

(Internet searching subscale) Search the Internet for images or photos on any device.

(Video gaming subscale) Play games on a computer, video game console or smartphone BY YOURSELF.

(Video Gaming Subscale) Play games on a computer, video game console or smartphone WITH OTHER PEOPLE IN THE SAME ROOM.

(Video gaming subscale) Play games on a computer, video game console or smartphone WITH OTHER PEOPLE ONLINE.

Which social media network do you use most frequently?

Facebook

Instagram

Twitter

Other (please specify) _____

How often do you do each of the following activities?

(General social media subscale) Check your social media.

(General social media subscale) Check your social media from your smartphone.

(General social media subscale) Check your social media at work or school.

(General social media subscale) Post status updates.

(General social media subscale) Post photos.

(General social media subscale) Browse profiles and photos.

(General social media subscale) Read postings.

(General social media subscale) Comment on postings, status updates, photos, etc.

(General social media subscale) Click "Like" to a posting, photo, etc.

Please answer the following questions about your Facebook and other online friends. NOTE: In items 41 and 42 the words "social media" (or any specific social media site) may be substituted for Facebook.

9-point scale for items 37–40 (with scoring in parentheses:

0 (1)

1–50 (2)

51–100 (3)

101–175 (4)

176–250 (5)

251–375 (6)

376–500 (7)

501–750 (8)

751 or more (9)

(Online friendships subscale) How many friends do you have on Facebook?

(Online friendships subscale) How many of your Facebook friends do you know in person?

(Online friendships subscale) How many people have you met online that you have never met in person?

(Online friendships subscale) How many people do you regularly interact with online that you have never met in person?

Attitudes. subscales

These subscales includes 16 items, which comprise four subscales: Positive Attitudes Toward Technology (6 items), Anxiety About Being Without Technology or Dependence on Technology (3 items), Negative Attitudes Toward Technology (3 items) and Preference for Task Switching (4 items)

5-point Likert scale for all items (with scoring in parentheses)

Strongly agree (5)

Agree (4)

Neither agree nor disagree (3)

Disagree (2)

Strongly disagree (1)

(Positive attitudes) I feel it is important to be able to find any information whenever I want online.

(Positive attitudes) I feel it is important to be able to access the Internet any time I want.

(Positive attitudes) I think it is important to keep up with the latest trends in technology.

(Anxiety/dependence) I get anxious when I don't have my cell phone.

(Anxiety/dependence) I get anxious when I don't have the Internet available to me.

(Anxiety/dependence) I am dependent on my technology.

(Positive attitudes) Technology will provide solutions to many of our problems.

(Positive attitudes) With technology anything is possible.

(Positive attitudes) I feel that I get more accomplished because of technology.

(Negative attitudes) New technology makes people waste too much time.

(Negative attitudes) New technology makes life more complicated.

(Negative attitudes) New technology makes people more isolated.

(Preference for task switching) I prefer to work on several projects in a day, rather than completing one project and then switching to another.

(Preference for task switching) When doing a number of assignments, I like to switch back and forth between them rather than do one at a time.

*(Preference for task switching) I like to finish one task completely before focusing on anything else.

(Preference for task switching) When I have a task to complete, I like to break it up by switching to other tasks intermittently.

*Scoring for item 15 is reversed with strongly agree = 1 and strongly disagree = 5.

Appendix C

Demographic Data

What age are you?

Which gender are you?

Male

Female

Other

Prefer not to say

What is your ethnicity?

White

Mixed or multiple ethnic group

Asian

African

Caribbean or Black

Other

Prefer not to say

What is your marital status?

Married

Widowed

Divorced

Separated

Never married

Other

Prefer not to say

What is your current living situation?

Alone

With spouse/partner

With family

In residential accommodation

Other _____

Prefer not say

What is the highest level of education you have completed?

Primary school

Secondary school

Higher education

College or university

Post-graduate degree

Prefer not say

What is your primary employment status?

Employed full time

Employed part time

Unemployed looking for work

Unemployed not looking for work

Retired

Student

Disabled

Prefer not to say

What is your household income?

Less than £10,000

£10,000 - £19,999

£20,000 - £29,999

£30,000 - £39,999

£40,000 - £49,999

£50,000 - £59,999

£60,000 - £69,999

£70,000 - £79,999

£80,000 - £89,999

£90,000 - £99,999

£100,000 - £149,999

More than £150,000

Prefer not to say

Appendix D

iPhone Data

This final section asks for information from your iPhone. To answer these questions go to Settings, then Screen Time, then select See All Activity and finally ensure you are looking at the Week tab. If

you need help finding screen time details please visit Apple support.

Enter the Daily Average screen time. This can be found at the top of the screen in large numbers.

Hours _____

Minutes _____

Go to See All Activity and select the week tab. Enter the time spent on Social Networking. This can be found directly under the first graph

Hours _____

Minutes _____

Scroll down to the Most Used section - what are your top 3 Most Used apps

Top _____

Second _____

Third _____

Scroll down to the Pickups section - enter your Daily Average number of pickups

Scroll down to the Notifications section - enter your Daily Average number of notifications

Appendix E

Stimuli Words

Appendix F

SPSS Output

FREQUENCIES VARIABLES=Gender Ethnicity MaritalStatus Education Employment Income
LivingSituation

/STATISTICS=STDDEV MEAN

/ORDER=ANALYSIS.

Frequencies

CORRELATIONS

/VARIABLES=calc_Stroop_AttentionalBiasResponseTime Age Gender Ethnicity MaritalStatus
Education

Employment Income LivingSituation

/PRINT=TWOTAIL NOSIG

/STATISTICS DESCRIPTIVES

/MISSING=PAIRWISE.

GET

FILE='/Users/juliehendry/Dropbox/Uni/C8983 - Empirical project/Data/SPSS/Untitled2.sav'.

DATASET NAME DataSet2 WINDOW=FRONT.

DATASET ACTIVATE DataSet1.

DATASET CLOSE DataSet2.

CORRELATIONS

/VARIABLES=iPhoneDailyAvgScreenTimeTotalMins iPhoneNumberOfPickups
iPhoneNumberOfNotifications

MTUAS_UsageSmartphone_factor MTUAS_UsageSocialMedia_factor SUQ_SmartphoneUsage

SUQ_AbsentMindedSmartphoneUsage

/PRINT=TWOTAIL NOSIG

/STATISTICS DESCRIPTIVES

/MISSING=PAIRWISE.

Correlations

GLM calc_Stroop_meanCorrectRT_CD calc_Stroop_meanCorrectRT_SMN BY
group_MTUASSocialMedia

group_MTUASSmartphone

/WSFACTOR=Stimuli 2 Polynomial

/METHOD=SSTYPE(3)

/POSTHOC=group_MTUASSocialMedia group_MTUASSmartphone(BONFERRONI)

/EMMEANS=TABLES(group_MTUASSocialMedia) COMPARE ADJ(BONFERRONI)

/EMMEANS=TABLES(group_MTUASSmartphone) COMPARE ADJ(BONFERRONI)

/EMMEANS=TABLES(Stimuli) COMPARE ADJ(BONFERRONI)

/EMMEANS=TABLES(group_MTUASSocialMedia*group_MTUASSmartphone)

/EMMEANS=TABLES(group_MTUASSocialMedia*Stimuli)

/EMMEANS=TABLES(group_MTUASSmartphone*Stimuli)

/EMMEANS=TABLES(group_MTUASSocialMedia*group_MTUASSmartphone*Stimuli)

/PRINT=DESCRIPTIVE ETASQ HOMOGENEITY

/CRITERIA=ALPHA(.05)

/WSDESIGN=Stimuli

/DESIGN=group_MTUASSocialMedia group_MTUASSmartphone
group_MTUASSocialMedia*group_MTUASSmartphone.

General Linear Model

GLM calc_Stroop_meanCorrectRT_CD calc_Stroop_meanCorrectRT_SMN BY
group_MTUASSocialMedia

group_iPhoneMins

/WSFACTOR=Stimuli 2 Polynomial

/METHOD=SSTYPE(3)

/POSTHOC=group_MTUASSocialMedia(BONFERRONI)

/EMMEANS=TABLES(Stimuli) COMPARE ADJ(BONFERRONI)

/PRINT=DESCRIPTIVE ETASQ HOMOGENEITY

/CRITERIA=ALPHA(.05)

/WSDESIGN=Stimuli

/DESIGN=group_MTUASSocialMedia
group_MTUASSocialMedia*group_iPhoneMins.

group_iPhoneMins

General Linear Model

Estimated Marginal Means

Stimuli

Post Hoc Tests

Social Media Use

CORRELATIONS

/VARIABLES=calc_Stroop_AttentionalBiasResponseTime Age

/PRINT=TWOTAIL NOSIG

/STATISTICS DESCRIPTIVES

/MISSING=PAIRWISE.

Correlations

CROSSTABS

/TABLES=calc_Stroop_AttentionalBiasResponseTime BY Gender Ethnicity MaritalStatus Education

Employment Income LivingSituation

/FORMAT=AVALUE TABLES

/STATISTICS=CHISQ PHI

/CELLS=COUNT EXPECTED ROW COLUMN TOTAL

/COUNT ROUND CELL.